Transformations and Norms

Psychological tests involve obtaining samples of behavior to measure psychological attributes such as;

- Intelligence
- Motivation
- Risk taking
- Psychopathology

The score on a scholastic test might reflect the number of questions answered correctly

The score obtained in an observation of mother-child interactions might be the number of times the mother smiles at her baby

The sort of behavior count is simple and concrete, but not necessarily informative

For example, suppose that you knew that:

- A student had answered 27 questions correctly on a statistics exam.
- A mother had smiled at her baby five times in 30 minutes.

Without additional information, it is impossible to determine whether:

- The student performed well.
- The mother was showing unusual warmth.

Raw scores represent simple counts of behaviors sampled by a test or measurement procedure.

However, raw scores **do not necessarily provide meaningful information**.

Why?

Because a raw score:

- Lacks context
- Does not indicate relative standing
- Does not allow comparison across individuals or test

It is often necessary to:
Re-express, or transform, raw scores into a more informative scale.

We use the term *scale* to refer to the set of scores that might be reported for a test

For example, scores on a test of general mental ability might be reported on an IQ scale

Scales are very often transformations of the original raw scores

The goal is to discuss the most widely used methods of transforming raw scores into more interpretable, and therefore more useful scales

The simplest transformation is one that expresses scores in terms of percentage rather than raw units

For example, if a student answered 27 questions correctly in a 40-item test, student's score could be reexpressed (transformed) as a score of 67.5%

This simple transformation has the three characteristics that many of the most useful transformation exhibit:

- It does not change the person's score; it is simply expresses that score in different units
- It takes into account information not contained in the raw score itself – here, the number of items on the test
- It presents the person's score in units that are more informative or interpretable than the raw score

Two different types of transformations:

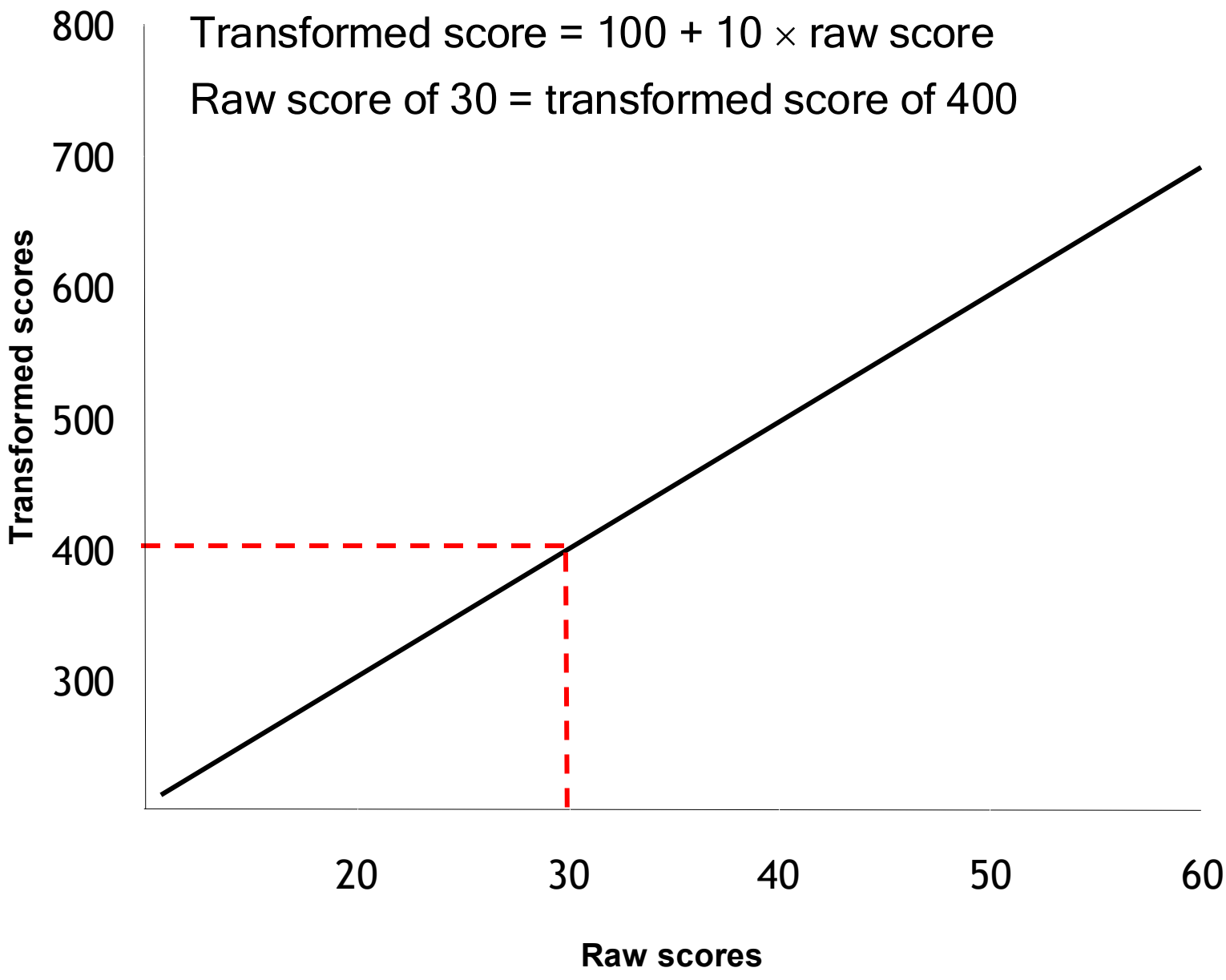
- Linear transformations
- Area transformations

A linear transformation is one that takes the form shown in equation:

$$\text{Transformed score} = \text{constant} + (\text{weight} \times \text{raw score})$$

This equation has the same general form as the regression equation

This transformation guarantess a direct relationship between raw scores and transformed scores that can be depicted using a straight line



One of the most familiar linear transformation is the <u>*z* score</u>

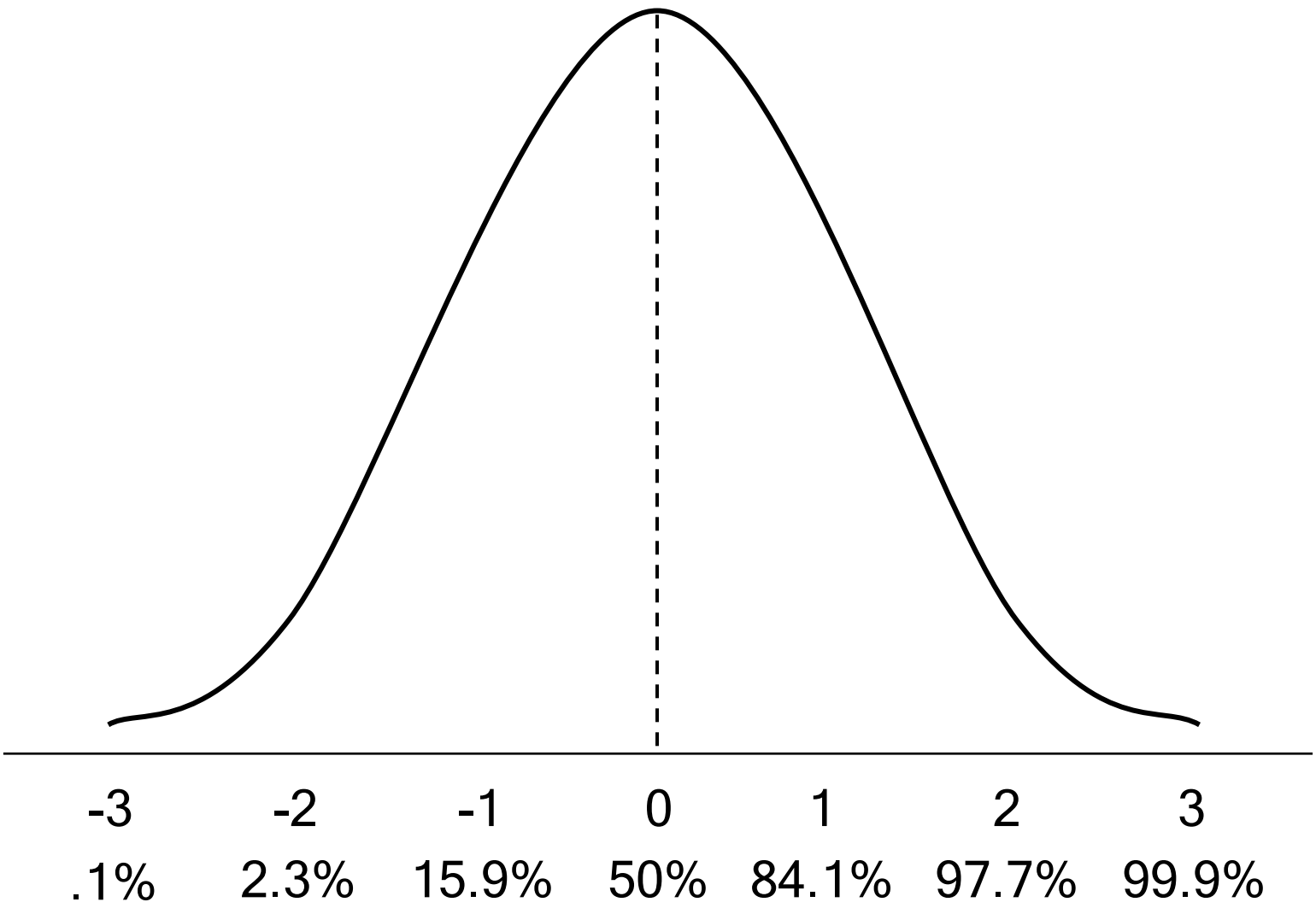
$$z = \frac{X - M}{SD}$$

z scores have a mean of 0 and standard deviation of 1; thus, an effective range of -3 to +3

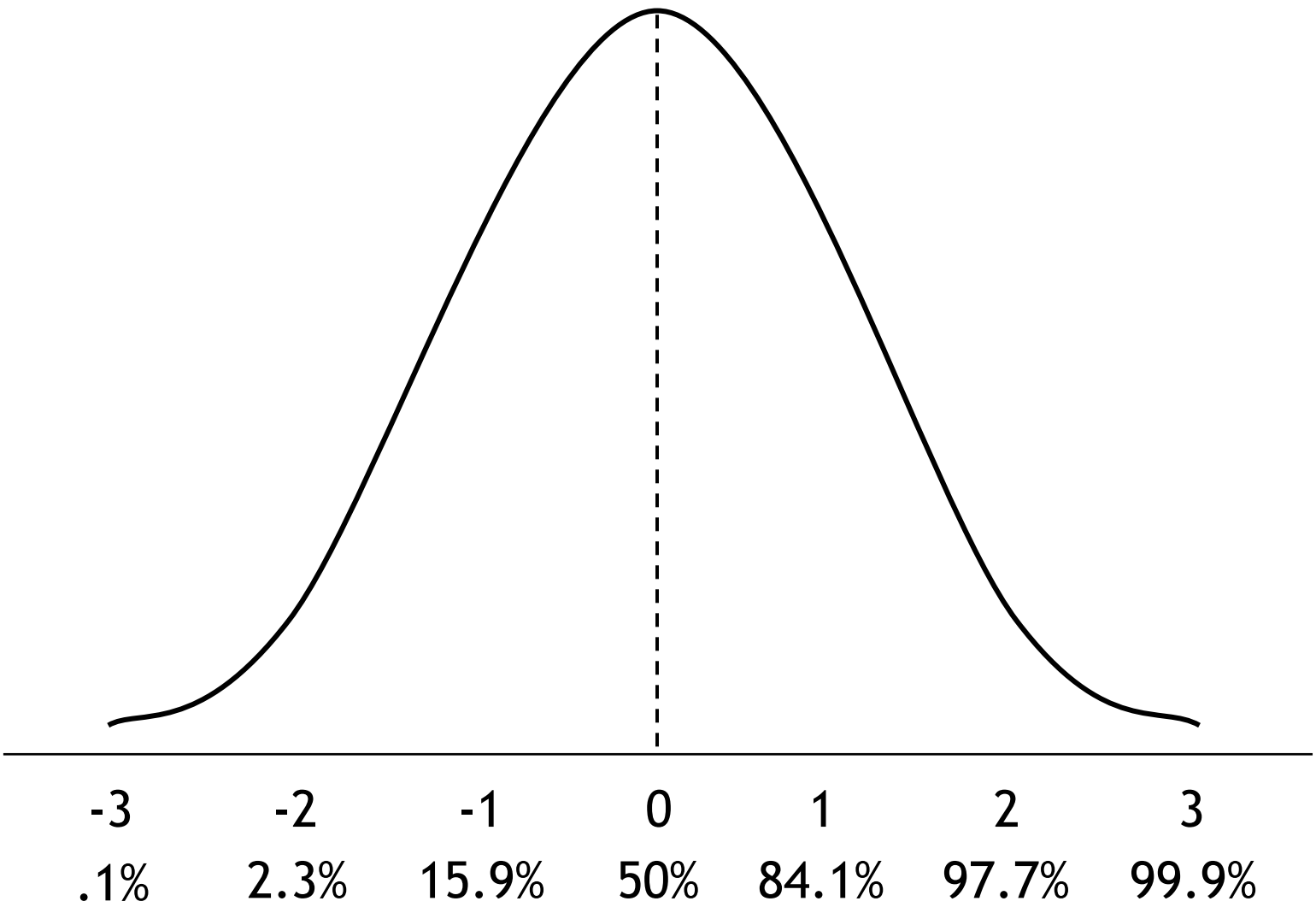
The z transformation is extremely useful for a number of reasons:

- The z scores indicates each person's standing as compared with group mean
- Those with positive z scores received raw scores that were above the mean, whereas those with negative z scores that below the mean
- When the distribution of raw scores is reasonably normal, z scores can be directly converted to percentiles

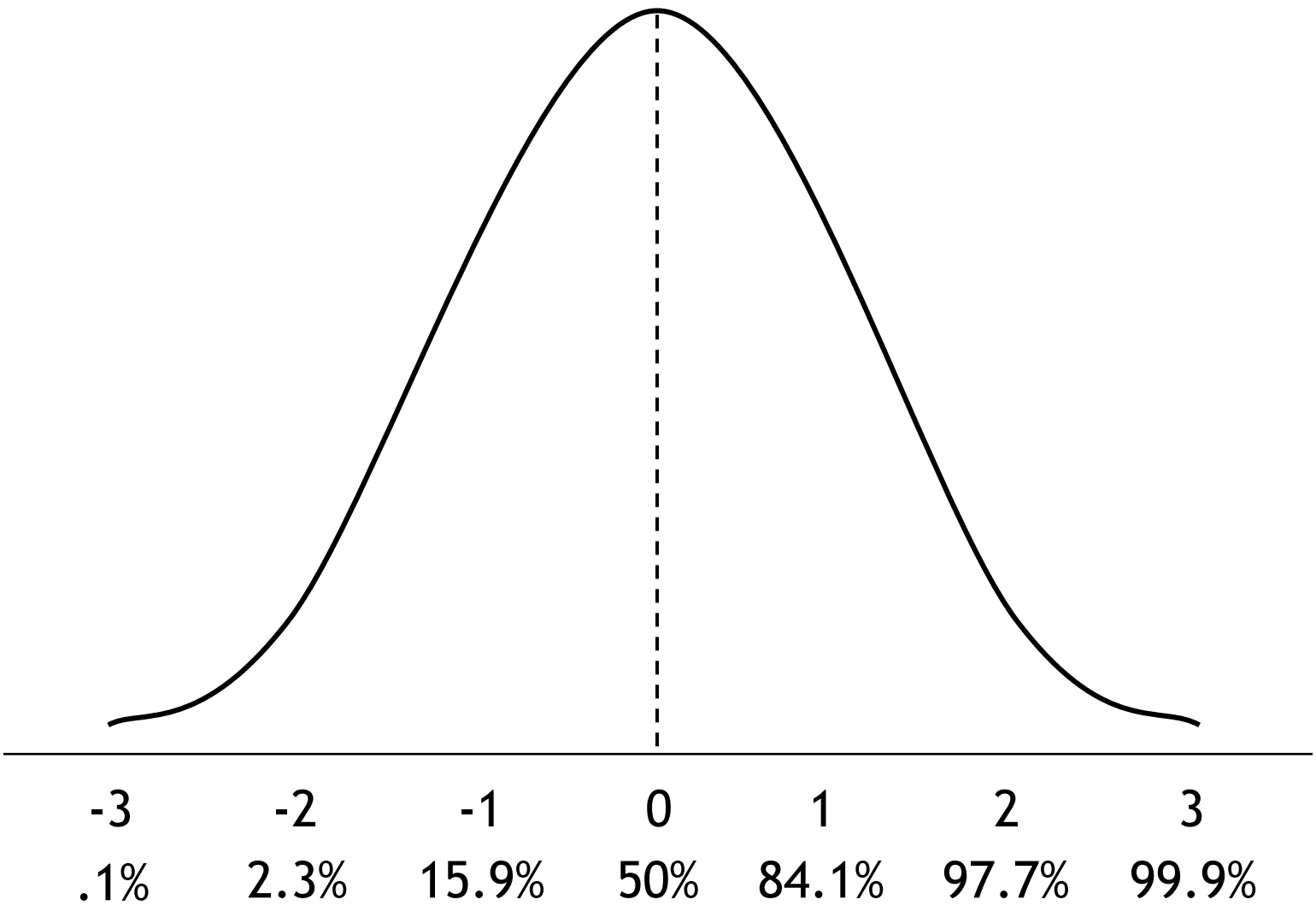
Figure illustrates the relationship between *z* scores and percentiles for the normal distribution



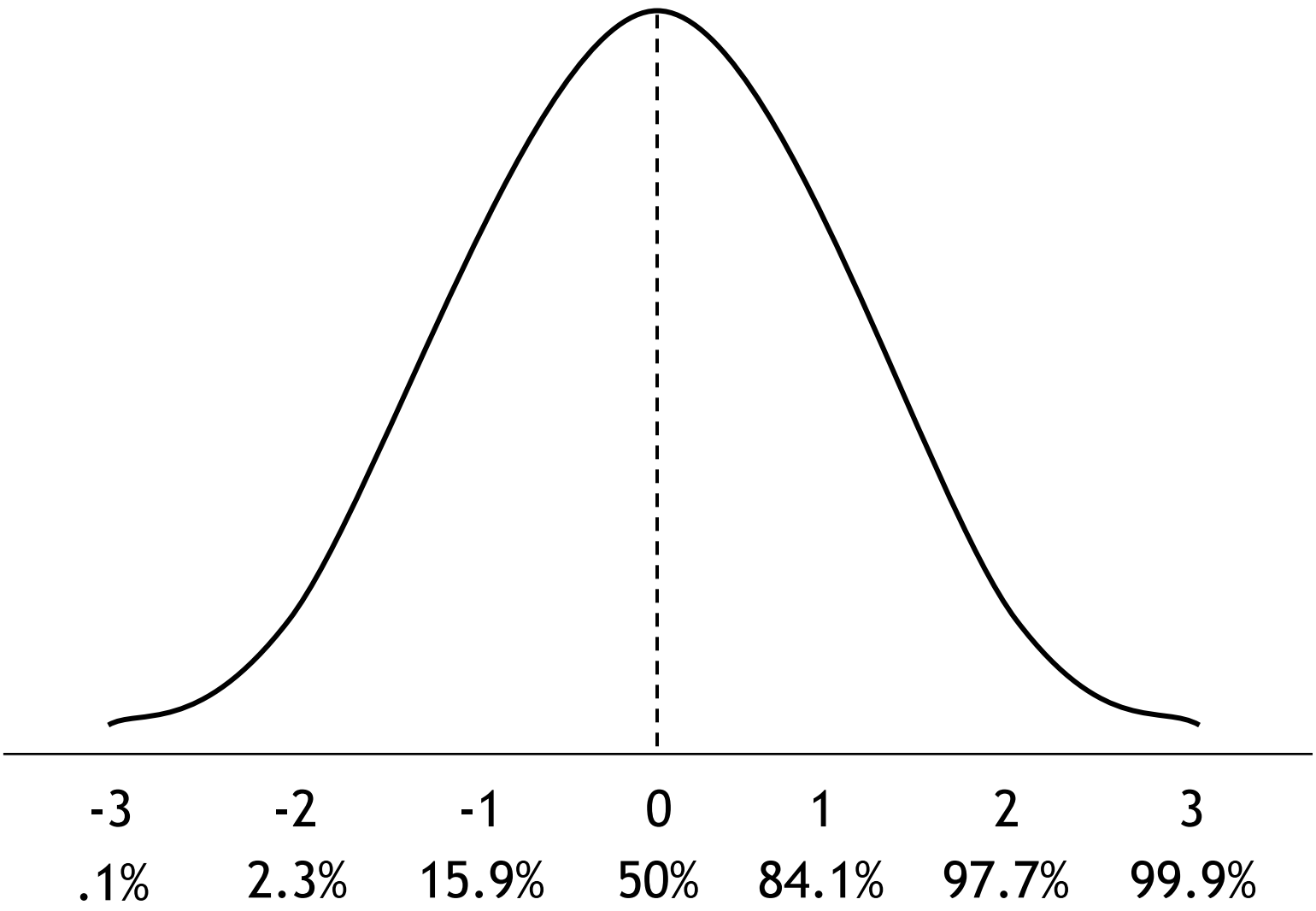
For example, 84.1% of the scores in a normal distribution are below a *z* score of 1.0



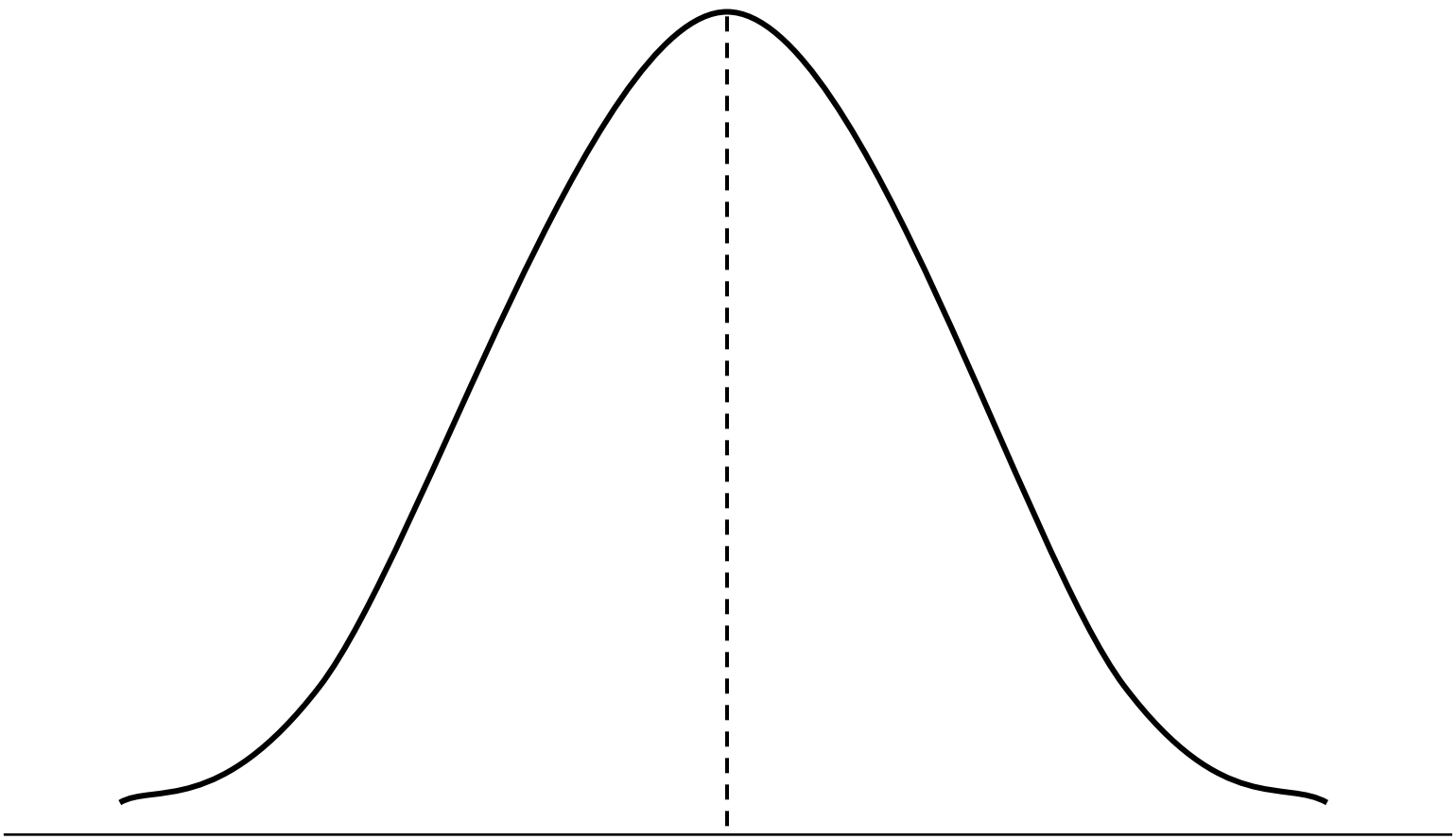
Therefore, if you received a *z* score of 1.0, you could conclude that you received a better score than 84% of the people taking test



Similarly, a *z* score of zero indicates that you did better than half of the people taking the test



This interpretation of *z* scores is accurate only when scores are normally distributed !!!



Variations of the *z* Transformation

There are two features of z scores that many people regard as undesirable

- Half of the *z* scores in a distribution will be negative
- *z* scores often have fractional values

Several score transformations can be used to overcome these two disadvantages

<u>T scores</u> illustrate this type of transformation

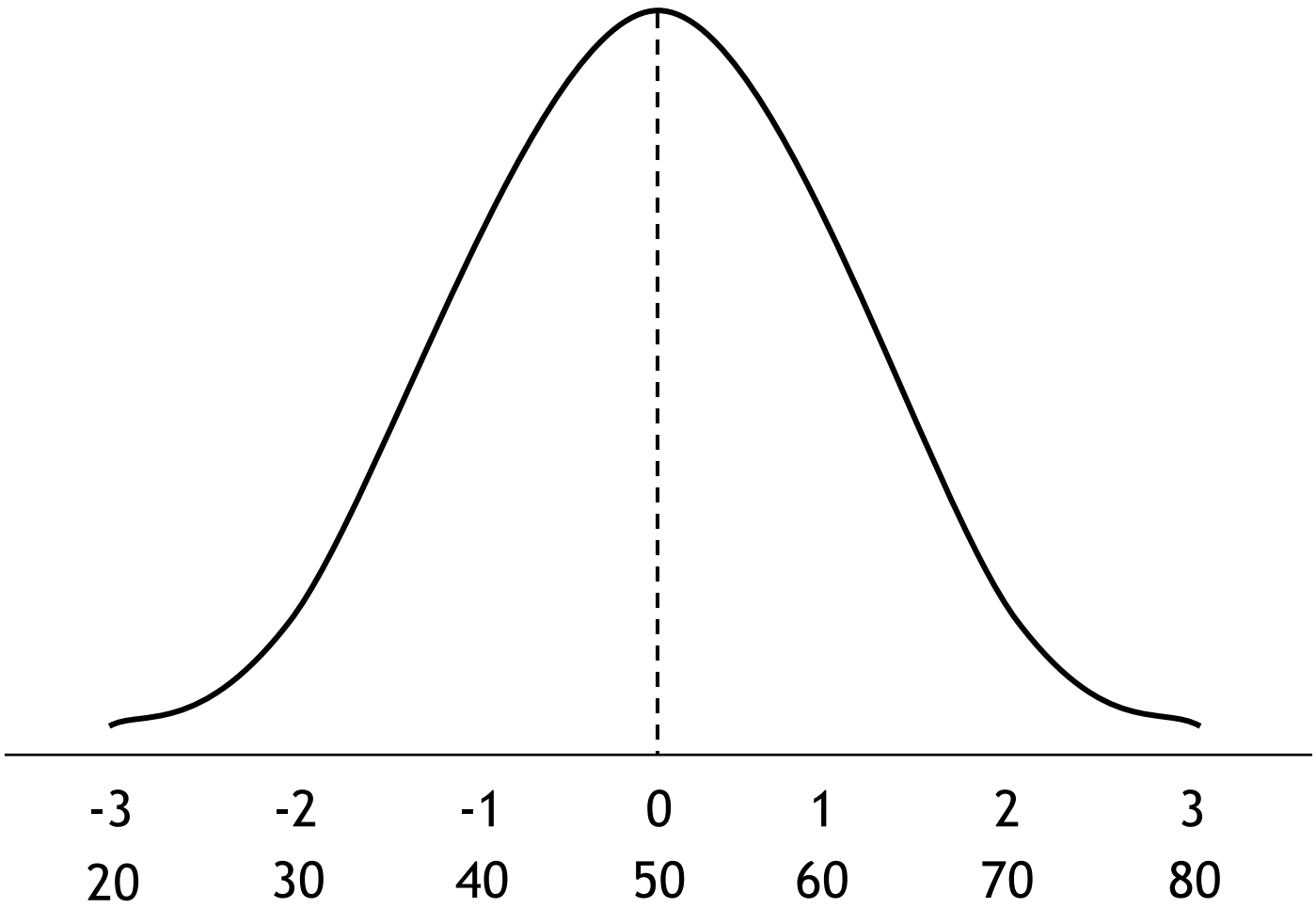
T score is a simple transformation of z that takes the following form:

$$\text{T score} = (\text{z score} \times 10) + 50$$

T scores have a mean of 50 and a standard deviation of 10

Thus, have an effective range of 20 to 80

Figure illustrates the relationship between *z* scores and T scores for the normal distribution



Area Transformations

- A second type of transformation, referred to as an area transformation, uses the normal curve to create scale score values
- Linear transformation simply change the units or measurement
- Area transformation are a bit more complex, because they do more than simply put scores on a new and more convenient scale

Rather than simply changing the units of measurement, area transformations change the point of reference

That is, these transformations express a person's score in terms of

- where it falls on a normal curve
- rather than simply providing a new unit of measurement

We have already mentioned one such area transformation, the percentile, or percentile rank

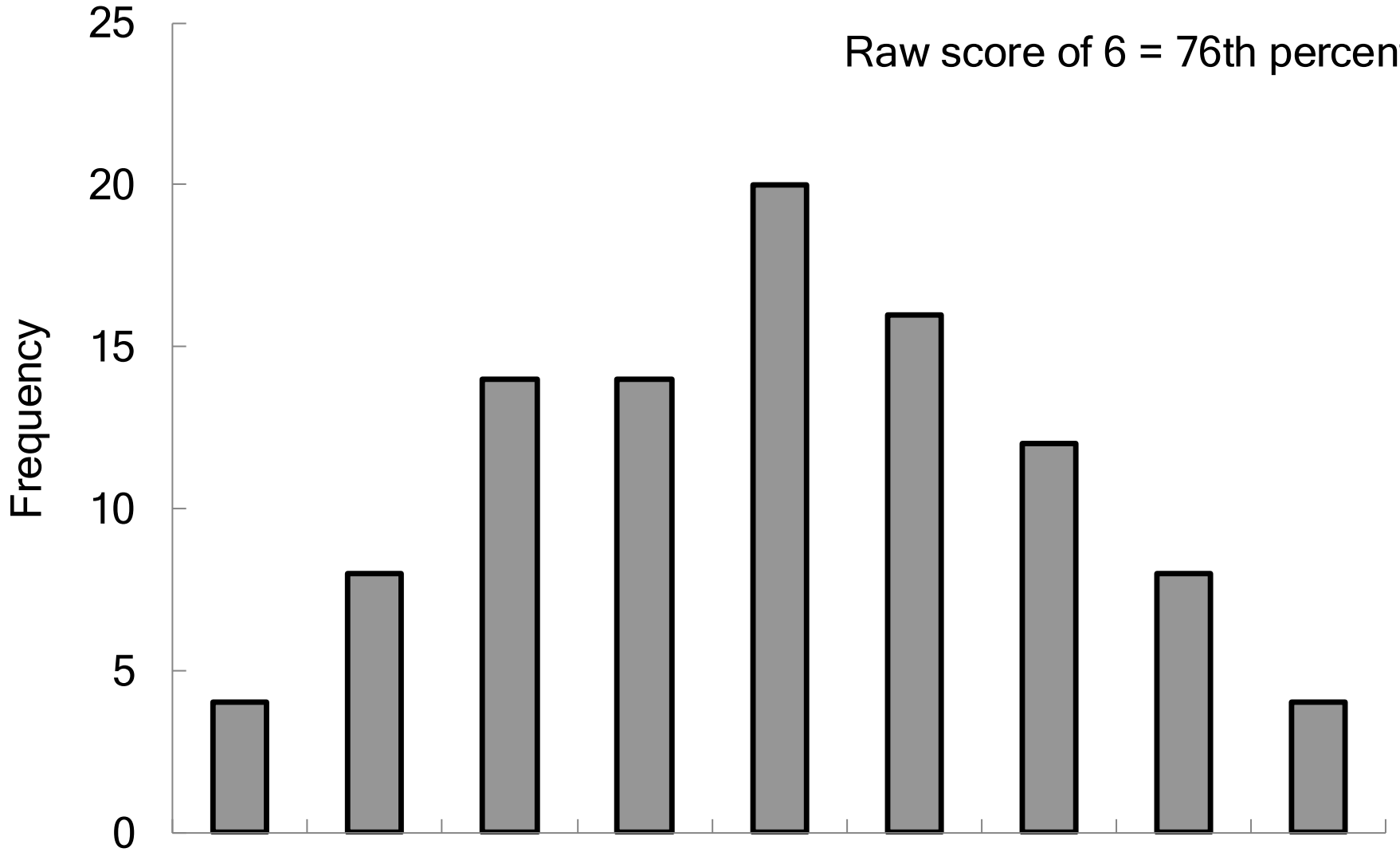
To show

- how this transformation relates to the concept of area
- how it changes the meaning of the test score

Following figure shows an examle of a reasonable normal distribution of 100 test scores expressed in raw scores and percentiles

Raw score of 3 = 26th percentile

Raw score of 6 = 76th percentile



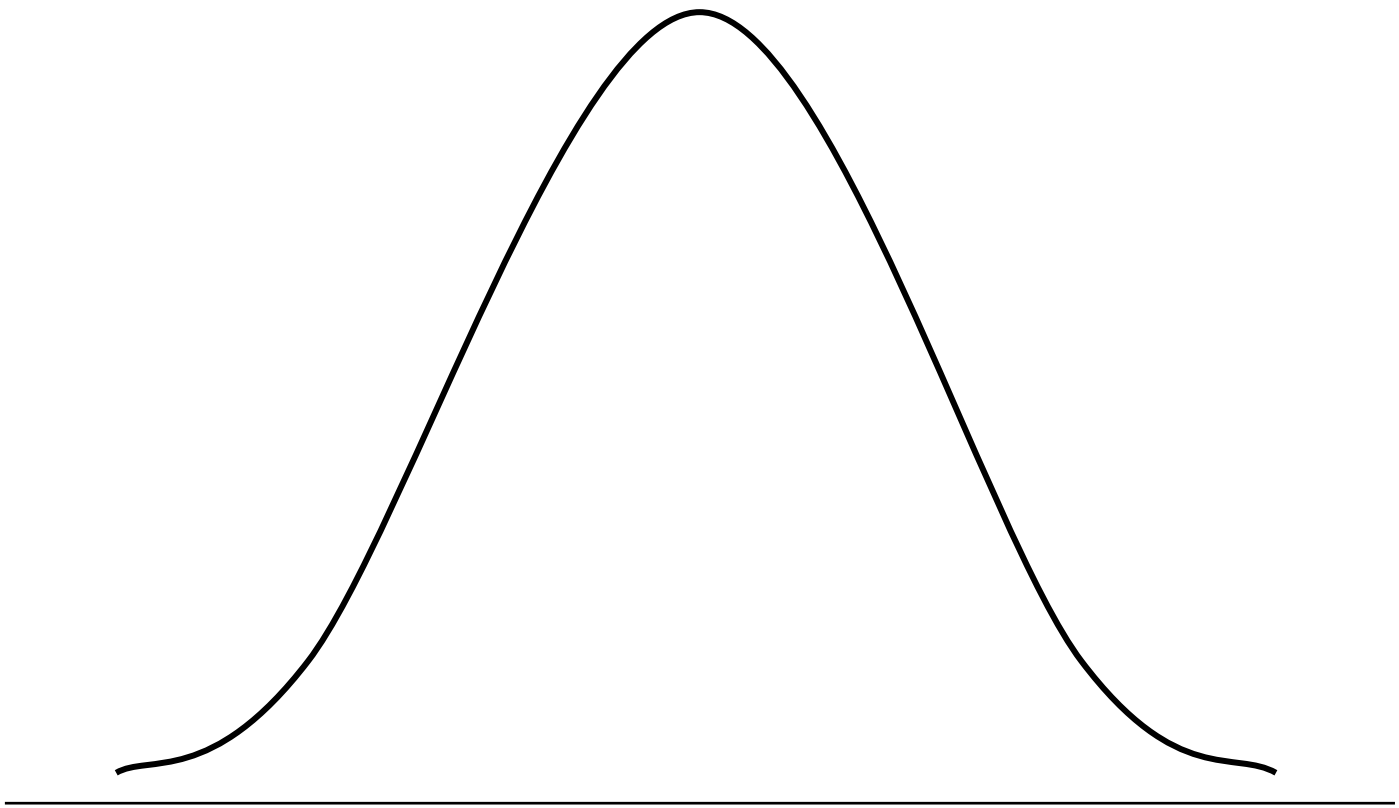
Raw score	1	2	3	4	5	6	7	8	9
Frequency	4	8	14	14	20	16	12	8	4
Percentage	4%	8%	14%	14%	20%	16%	12%	8%	4%
Percentile	4%	12%	26%	40%	60%	76%	88%	96%	99.9%

- Notice how the percentile transformation changes the relationship between test scores
- The difference between a test score of 1 and a score of 2 is the same as the difference between a 5 and a 6
- The percentiles, however, differ considerably
- Scores near the center of the distribution differ substantially in their percentile ranks
  - 5 = 60% and 6 = 76%
- Whereas scores at the extremes have more similar percentile ranks
  - 1 = 4% and 2 = 12%

- This percentile transformation magnifies differences between individuals at the middle of the distribution and compresses the differences between individuals at the extremes
- Area transformations do not change the rank order of scores, but they do change the intervals between scores

- As we have mentioned, scores on many psychological tests are likely to reflect ordinal rather than interval measurement scales,
- So, the fact that area transformations can change intervals between scores is probably not a major problem

Figure illustrates percentile area transformation in relation to a normal curve, and it helps to clarify the way in which the concept of area is used to transform raw data



z score	-3	-2	-1	0	1	2	3
T score	20	30	40	50	60	70	80
Percentile	1%	2%	16%	50%	84%	98%	99.9%

Raw score	z score	T score	Percentile
70			
80			
90			
100			
110			
120			
130			

The data in this table are from a test in which scores are normally distributed, with a mean of 100 and standard deviation of 15

Raw score	z score	T score	Percentile
70	-2.0		
80	-1.33		
90	-.67		
100	0		
110	.67		
120	1.33		
130	2.0		

Raw score	z score	T score	Percentile
70	-2.0	30	
80	-1.33	37	
90	-.67	43	
100	0	50	
110	.67	57	
120	1.33	63	
130	2.0	70	

Raw score	z score	T score	Percentile
70	-2.0	30	2.3
80	-1.33	37	9.2
90	-.67	43	25.1
100	0	50	50
110	.67	57	74.9
120	1.33	63	90.8
130	2.0	70	97.7

- Raw score of 80 = percentile rank of 9.2
  - 9.2% of all scores are 80 or lower
- Over 90% of the scores are 120 or lower

Norms

- Scores on psychological tests rarely provide absolute, ratio scale measures of psychological attributes
- Thus, it rarely makes sense to ask, in an absolute sense, how much intelligence, motivation, depth perception, and so on, a person has

- Scores on psychological tests do, however, provide useful relative measures
- It makes perfect sense to ask whether Ahmet is more intelligent, is more motivated, or has better depth perception than Tuna
- Psychological tests provide a systematic method of answering such questions

- One of the most useful ways of describing a person's performance on a test is to compare his or her tests score with the test scores of some other person or group of people
- Many psychological tests base their scores on a comparison between each examinee and some standard population that has already taken test
- The YKS, in which the scores of each individual are compared with the scores of thousands of other high-school students, provides an example of this approach

- When a person's test score is interpreted by comparing that score with the scores of several people, this comparison is referred to as a *norm-based interpretation*
- The scores with which each individual is compared are referred to as norms, which provide standards for interpreting test scores
- A norm-based score indicates where an individual stands in comparison with the particular normative group that defines the set of standards

- Several different groups might be used in providing normative information for interpreting test scores
- For example, the raw score of an entry-level secretary on a typing speed test could be compared with,
  - The scores of people in general, such as might be obtained in a random census survey
  - The scores of experienced typists
  - The scores of other entry-level secretaries

Types of Norms

- In some case, norms may be developed that are national in scope, as in the case of large achievement batteries
- In these instances, norm groups may involve many thousands of children, and tens of thousands of individuals may take the test annually

- In the development of local norms to be used in much smaller testing applications, the practical problems are considerably less complex
- Although the logistics of compiling local norms are typically less complex than those involved when compiling national norms, the basic process is not really different

- Normative data can be expressed in many ways
- Three of the most common forms are
  - Percentile ranks
  - Age norms
  - Grade norms

Percentile Ranks

- The most common form of norms is percentile rank
- The percentile rank represents the percentage of the norm group that earned a raw score less than or equal to the score of that particular individual
- It is possible to compare one's score with several different norm groups

- The table in the following slide is an adaptation taken from the Norms Booklet for the Clerical Speed and Accuracy test of the Differential Aptitude Tests
- The purpose of the test is to see how fast and accurate examinees can be in a 100-item checking task
- The score is simply the number of items that are correctly checked in 3 minutes
- The norms table is based on both girl and boys who were tested at the beginning of grades 8 through 11

Percentile	Grade 8		Grade 9		Grade 10		Grade 11		Percentile
	B Fall	G Fall	B Fall	G Fall	B Fall	G Fall	B Fall	G Fall	
99	58-100	63-100	62-100	67-100	66-100	73-100	69-100	78-100	99
97	54-57	58-62	56-61	62-66	59-65	65-72	61-68	69-77	97
95	49-53	54-57	52-55	58-61	54-58	61-64	55-60	63-68	95
90	46-48	52-53	49-51	55-57	51-53	58-60	53-54	60-62	90
85	44-45	50-51	47-48	53-54	49-50	56-57	51-52	58-59	85
80	43	48-49	45-46	51-52	48	54-55	49-50	56-57	80
75	42	47	44	49-50	47	52-53	48	54-55	75
70	40-41	46	43	48	46	50-51	47	53	70
65	39	45	42	46-47	45	49	46	52	65
60	38	44	40-41	45	44	48	45	51	60
55	37	43	39	44	43	47	44	50	55
50	36	42	38	43	42	46	43	49	50
45	35	41	37	42	41	45	42	47-48	45
40	34	39-40	36	41	39-40	44	41	46	40
35	32-33	38	35	40	38	43	40	45	35
30	31	36-37	33-34	39	37	42	39	44	30
25	29-30	34-35	32	38	35-36	40-41	37-38	42-43	25
20	28	32-33	30-31	36-37	33-34	38-39	35-36	40-41	20
15	26-27	30-31	28-29	32-35	31-32	36-37	32-34	38-39	15
10	23-25	26-29	25-27	28-31	28-30	32-35	30-31	35-37	10
5	19-22	22-25	21-24	24-27	23-27	26-31	25-29	30-34	5
3	13-18	16-21	14-20	19-23	18-22	22-25	20-24	25-29	3
1	0-12	0-15	0-13	0-18	0-17	0-21	0-19	0-24	1

- Several interesting conclusions can be drawn from these data
  - The data show a general increase in performance throughout the grades
  - The very lowest percentile ranks have raw scores that could be classified as in or close to the chance range

    Since this test has 100 items, each having 5 options, a chance score would be close to 20

    One therefore would have to interpret the lower percentile ranks very carefully
  - Perhaps most important, notice that often the scores for girls must be several points higher to achieve the same percentile rank as boys

    Girls performed better on this test than the boys

Age Norms

- Many psychological characteristics change over time; vocabulary, mathematical ability, and moral reasoning are examples
- Although there may be a general pattern of growth for these types of variables, often there are wide variations among people as to whether they grow at faster or slower rates

- An age norm relates a level of test performance to the age of people who have taken test
- For example, one condition of the Stanford-Binet Intelligence Test has a vocabulary subtest that consist of 46 items, arrange from the easiest to the most difficult
- A score for examinee represents the number of words defined correctly

Age level	Test score
14	31
12	28
10	25
8	22
6	18

Table presents some normative data from this particular vocabulary test

For example, for the typical child at:

- age 6, the average number of correct definitions is about 18
- age 12, the number is about 28

- The principle involved in developing age norms is fairly simple:
- They can be developed for any characteristics that changes systematically with age – at least up to some age level
- In establishing age norms, we need to obtain a representative sample at each of several ages and to measure the particular age-related characteristic in each of these samples

Grade Norms

- Another type of norm commonly used in school settings is called a grade norm
- Grade norms are very similar to age norms except that
- The baseline on the graph is the grade level rather than age
- These norms are most popular when reporting the achievement levels of school children
- The interpretation of grade norms is similar to that for age norms

- The table in the following slide shows data from the California Achievement Tests, in particular, the vocabulary subtest
- The vocabulary subtest has two parts of 20 items each, and scores of these two parts are summed together
- Thus, scores on this test could range from zero to 40

Grade	*VOCAB*	*Grade*	*VOCAB*	*Grade*	*VOCAB*
0.6	0-12	2.8		5.0	
0.7	13	2.9	32	5.1	
0.8		3.0	33	5.2	
0.9	14	3.1	34	5.3	39
1.0		3.2		5.4	
1.1	15	3.3	35	5.5	
1.2		3.4		5.6	
1.3	16	3.5		5.7	
1.4	17	3.6	36	5.8	
1.5		3.7		5.9	
1.6	18	3.8		6.0	
1.7	19	3.9		6.1	
1.8	20	4.0	37	6.2	
1.9	21	4.1		6.3	
2.0	22	4.2		6.4	40
2.1	23-24	4.3		6.5	
2.2	25	4.4		6.6	
2.3	26	4.5		6.7	
2.4	27	4.6	38	6.8	
2.5	28	4.7		6.9	
2.6	29	4.8		7.0	
2.7	30-31	4.9		7.1	

The median score for children just beginning the second grade is about 22, whereas the typical score for a beginning fourth grader is approximately 37

- What happens if a student who is just beginning grade 2 earns a scale score of 37 in this test?
- In this case, the child would receive a grade-equivalent score of 4.0.
- Does this score mean that;
- This child who is beginning second grade is achieving like a typical child who is beginning fourth grade?
- The answer to this question is both yes and no

- One needs to be extremely careful when interpreting grade norms not to fall into the trap of saying that just because a child obtains a certain grade-equivalent value on a particular test, he or she is like a child of that grade-equivalent level in all areas
- In areas such as emotional and social growth, as well as in other achievement areas, the child may not perform at that grade equivalent

Cautions for Interpreting Norms

- Normative information can be misinterpreted in many ways
- These range from
  - using normative data that were based on inappropriate target populations to
  - simply misinterpreting the particular statistics underlying normative data

- The most important of these potential sources of misinterpretation occurs when the norms provided in the test manual or in other types of articles are not based on samples that adequately represent the type of population to which the examinee's scores are to be compared
- A careful reading of the technical manual generally assists in deciding whether the norms are at least close to those desired
- However, many times even these statements can be misleading
- One way to help to resolve this problem is to develop local norms on samples for which one has control over selection and testing

- A second difficulty in interpretation that was mentioned in reference to grade norms should be reemphasized here
- Grade norms, and to some extent age norms, can easily be seen as saying that the examinee should be performing at the tested level not only of that specific test but also in other areas and behaviors
- It is fairly easy to assume mistakenly that a grade of 7.5 a student who is actually in the latter part of the eight grade means that he or she is capable of acting, in all respects, only like a typical student in the middle of the seventh grade

- Another problem with normative data is that they can become out of date very quickly
- Since the process of developing and norming a test is very time-consuming and expensive, most large-scale tests are not renormed very often
- This situation is also particularly true for less well-known instruments
- Not only should norms be based on samples that represent adequately the target population, but they also should be current

- An additional problem that may be encountered with normative data has to do with the size of the sample taken
- Typically, when we look at the norms, there is a tendency to assume that they are fixed and not subject to error
- That is, the temptation to think that if the same norm group were reexamined with the same test, their score would produce the same pattern as expressed in the norms manual

- However, if norm groups are small, there may be considerable sampling error
- That is, is another group similar to the norm were tested to renorm the test, the distributions and data derived from the new group would not necessarily be the same as the first
- If the size of the original group is sufficiently large, changes in norms due to sampling error are generally too small to make a practical difference